

Water Vapor Robustness of Hollow-Core Fiber for O-Band Short Reach Interconnects

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Abstract *This work presents the first experimental investigation of water vapor's impact on O-band hollow-core fibres. Results show that HCFs are robust to water vapor: even after prolonged high-humidity exposure, attenuation at the O-band operating wavelengths remains virtually unaffected, confirming their reliability for O-band short-reach interconnects.* ©2026 The Author(s)

Introduction

The proliferation of machine learning applications driven by large language models (LLMs) is creating exponential growth in intra- and inter-data centre transmission demands [1,2]. Efficient LLM training requires low-latency, high-bandwidth interconnections across massive computing clusters, where even nanosecond-level latency accumulation during GPU synchronization significantly impacts efficiency. Conventional optical interconnects using standard single-mode fiber (SMF) face pronounced latency limitations in ultra-high-density, short-reach scenarios.

Hollow core fibres (HCFs) have emerged as a promising alternative, guiding light in an air-filled core via an anti-resonant mechanism [3-6]. This enables superior properties such as significantly reduced latency and ultra-low attenuation, making HCFs attractive for data centre interconnects. However, gas absorption [7-10]—particularly from water vapor (H_2O) in the O-band—poses a potential challenge for short-reach applications, as atmospheric gases can enter the core after fabrication.

This work presents the first experimental investigation of water vapor's impact on O-band HCF transmission. We quantify the evolution of the vapor-absorption spectrum under high humidity and conduct real-time 400G-FR4 transmission tests. Results indicate that vapor ingress introduces absorption lines in the O-band but does not degrade the attenuation at operational wavelengths, even at high concentrations, demonstrating the HCF's robustness to water vapor. 400G-FR4 transmission results confirm that HCFs with high vapor concentrations perform comparably to solid-core single-mode fibres.

Characterization and Impact of Water Vapor

Absorption

According to the gas absorption data provided by the HITRAN database, water vapor introduces absorption loss in the O-band (1260–1360 nm), E-band (1360–1460 nm), and S-band (1460–1530 nm) [11,12]. The absorption lines of water vapor overlap with the operating wavelengths specified in ITU-T G.694.2 and widely adopted in pluggable optical modules for O-band CWDM, namely 1271 nm, 1291 nm, 1311 nm, and 1331 nm. Water vapor absorption introduces additional loss at these operating wavelengths when the total vapor concentration is sufficiently high, posing a potential risk to HCFs in short-reach O-band interconnects.

To characterize HCF attenuation under water vapor ingress, a high-precision insertion loss (IL) test system paired with an environmental chamber was employed, which is shown in Fig. 1(a). The HCF sample was coupled at both ends to single-mode fibres connected to a swept-wavelength laser (EXFO T500S, 1 pm resolution) and an IL test platform (EXFO CTP10). To enable clear observation of absorption changes, the coupling points (which were not hermetically sealed) were placed in an 85 °C/85 % RH chamber, allowing controlled ingress of water vapor. The HCF samples were the Gap Tube Assisted Support-Tube Hollow-Core Fiber (GTA-ST-HCF) [13] with low loss and low Intermodal Interference. The end-face microstructure is shown in Fig. 1(b).

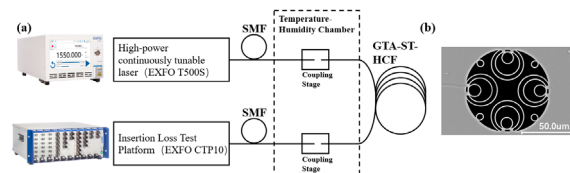


Fig. 1: (a) Schematic diagram of the IL test setup, (b) micrograph of the fiber end-face.

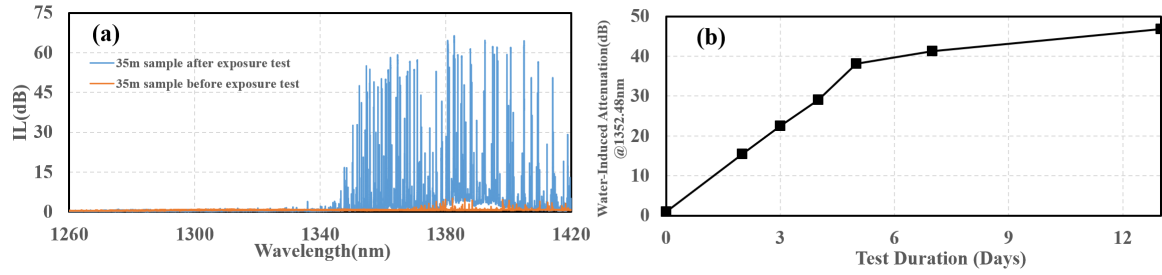


Fig. 2: (a) Temporal evolution of the gas absorption spectrum for the 35m GTA-ST-HCF sample under 85 °C/85 % RH conditions; (b) water-vapor-induced attenuation at 1352.48 nm over time.

Based on the aforementioned IL test setup, we conducted a series of water vapor exposure test to analyse the impact of water vapor on the attenuation spectrum of the O-band GTA-ST-HCF. We firstly measured the temporal variation of the gas absorption spectrum of a 35-meter O band GTA-ST-HCF over the 1260 nm to 1420 nm range after exposure in the environmental chamber. The time dependent evolution of the gas absorption spectrum of the GTA-ST-HCF sample under 85°C/85% RH steady-state test condition is shown in Fig. 2.

Fig. 2(a) shows the initial water vapor absorption spectrum, where absorption lines in the 1350-1420 nm band exhibited relatively low intensity. After exposure to high temperature and humidity, the absorption increased significantly. Specifically, the absorption line centered at 1352.48 nm was chosen to represent the overall absorption intensity, owing to its moderate strength and isolation from adjacent peaks. The temporal evolution of the additional loss introduced by the absorption line at 1352.48 nm is shown in Fig. 2(b). It is clear that the additional loss at 1352.48 nm increased rapidly after the first 5 days exposure, rising from 0.96 dB to 38.1 dB. Subsequently, the growth rate slowed significantly. Between day 5 and day 13, the additional loss increased from 38.10 dB to 46.85 dB.

The observed trend is consistent with the gas dynamic of water vapor ingress in hollow-core fibres [8]. Water vapor enters the fiber core in two sequential phases: an initial rapid influx driven by the pressure differential, followed by a slower, diffusion-dominated phase after pressure equilibration.

Furthermore, after 13 days of water vapor exposure test, no measurable absorption lines were observed near the O-band operating wavelengths in the absorption spectrum of the 35-m sample. Given the 35 m sample's short length and low initial vapor concentration, its limited vapor capacity could not adequately assess whether absorption affects O-band operating wavelengths. Moreover, during coupling, internal pressure could rapidly equilibrate with the external environment, which altered the test pressure

differential and further reduced vapor ingress.

To compare the total water-vapor concentration under different fiber lengths, initial pressure distributions, and initial vapor concentrations, we performed vapor exposure tests on 500 m and 2 km GTA-ST-HCF samples. Having been exposed to ambient air and subjected to the ensuing gas ingress—driven by the pressure differential and accompanied by prolonged diffusion—both sample sets exhibited a relatively high initial water-vapor concentration prior to testing.

Fig. 3 shows the water-vapor absorption spectrum of the 500 m and 2 km samples before and after the exposure test. The initial spectrum re-

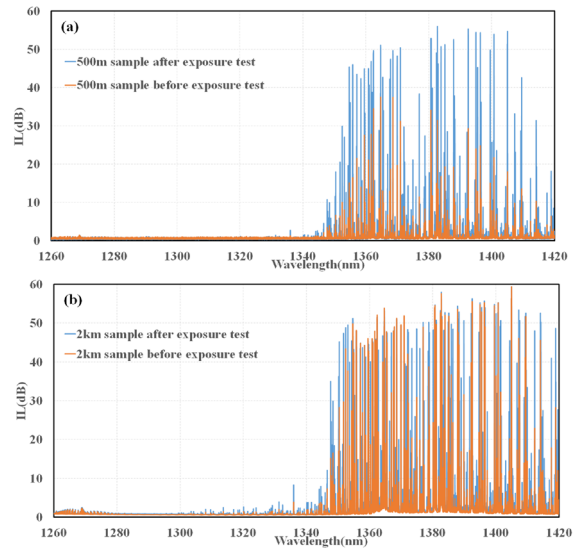


Fig. 3: Water-vapor absorption spectrum of hollow-core fiber samples before and after 10-day exposure test: (a) 500m; (b) 2 km sample.

vealed that both sample sets had a higher initial vapor concentration than the 35 m sample.

After 10 days of exposure test, the spectrum indicated a further increase in vapor concentration. Choosing the absorption line at 1352.48 nm as a reference, the additional loss of the 500 m sample increased from 9.39 dB to 47.75 dB after exposure, while that of the 2 km sample rose from

42.71 dB to 47.97 dB. It should be noted that, because water-peak attenuation in the 1350~1420 nm band approached the measurement limit of the IL test platform, the measured attenuation for some absorption lines no longer changed even though the overall vapor concentration continued to rise.

The 2 km sample already exhibited measurable water-peak loss in the 1300~1340 nm band. It should also be noted that, with increased fiber length, the absorption lines of oxygen in the 1260 nm to 1280 nm band accumulated to a measurable level. The IL of the 2 km GTA-ST-HCF and SMF samples at the operating O-band wavelengths during the exposure test are listed in Tab. 1. Although minor variations occurred due to measurement error, the IL at the operating wavelengths showed no measurable change before and after water vapor exposure test.

Tab. 1: The IL of the 2 km GTA-ST-HCF sample at the O-band operating wavelengths during the exposure test.

Wave-length (nm)	IL of GTA-ST-HCF (dB)		
	Initial State	after 5-day exposure test	after 10-day exposure test
1271	0.699	0.940	0.897
1291	0.699	0.841	0.769
1311	0.761	0.799	0.690
1331	0.639	0.814	0.698

Transmission performance

To assess whether high water vapor concentration in the core affects the transmission performance of the O-band GTA-ST-HCF, we compared the bit error rate (BER) performance of the 2 km GTA-ST-HCF sample previously subjected to water vapor exposure with a 2 km O-band SMF. The experiment set up is shown in Fig. (4).

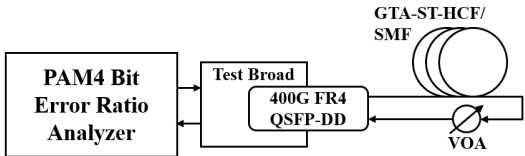


Fig. 4: BER Test Platform for Optical Transceivers

A real-time transceiver system based on the commercial 400G-FR4 optical module was employed. Variable optical attenuator (VOA) controlled the received optical power per channel, and a PAM4 BER analyzer combined with a test board measured the BER performance across the O-band channels.

The IL of the 2 m GTA-ST-HCF and SMF samples at the operational O-band wavelengths are listed in Tab. 2. The results show that, despite the high water vapor concentration in the HCF, its IL at these operating wavelengths is not measurably affected and remains comparable to that of

Tab. 2: Comparison of IL between the 2 km HCF and SMF samples at the O-band operating wavelengths.

Wave-length(nm)	IL of GTA-ST-HCF (dB)	IL of SMF (dB)
1271	0.897	0.858
1291	0.769	0.809
1311	0.690	0.720
1331	0.698	0.640

the SMF.

Fig.5 shows the BER performance for the four channels (1271 nm, 1291 nm, 1311 nm, 1331 nm) of the module under an operating temperature of 70 °C. For the 2 km GTA-ST-HCF sample containing water vapor, all four channels achieve transmission performance comparable to that of the SMF. The results confirmed that even at high water vapor concentrations, the GTA-ST-HCF achieved comparable transmission performance to conventional SMF, as the IL at its operating wavelengths remains unaffected.

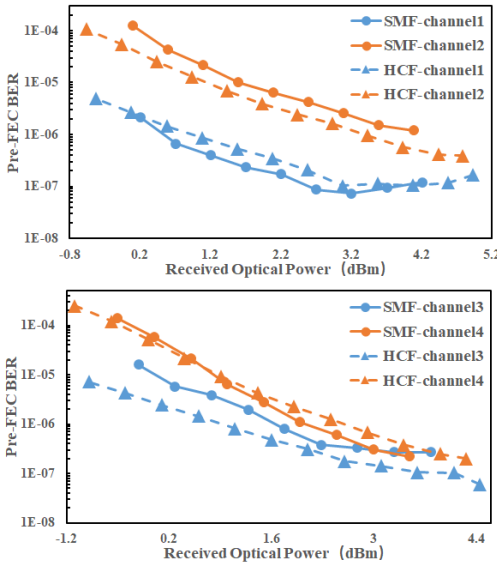


Fig. 5: BER performance for each channel of HCF and SMF at 70 °C.

Conclusion

This work experimentally investigates the impact of water vapor absorption on the spectral characteristics and transmission performance of O-band HCFs. Water vapor exposure tests result shows that O-band HCFs exhibit robustness to water vapor: even after prolonged exposure to ambient and high-temperature, high-humidity environments, the attenuation at the O-band operating wavelengths shows no significant degradation. 400G-FR4 transmission results confirm that HCFs with high vapor concentrations perform comparably to solid-core single-mode fibres.

Acknowledgements

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